

Description

On February 15, 2025 at 7:00 AM, an 8-week old infant is admitted to NIST Hospital with severe respiratory symptoms, including cough, fever, decreased appetite, wheezing and apnea. The infant's pediatrician has ordered the admission, suspecting the child may have contracted Respiratory Syncytial Virus (RSV). A clerical assistant registers the patient for admission. He records the patient's name, date of birth, race, ethnicity, residence, and insurance information. The triage nurse documents the admitting diagnoses as indicated by the pediatrician as:

56018004^Wheezing (finding)

386661006^Fever (finding)^SCT

724229002^Apnea of infancy (disorder)

700213005^Avoidant restrictive food intake disorder (disorder)

417850002^Respiratory tract congestion and cough (disorder)

The triage nurse also documents travel history indicated by the parents that they had spent 4 days at an amusement park and returned 8 days ago.

NIST Hospital reports inpatient syndromic surveillance data to the State Department of Public Health (NIST PH SS FAC 2) for all new patient admissions at least at least once per day. At 7:15 AM on February 15, 2025, the hospital's electronic health record module for syndromic surveillance data assembles and transmits an ADT^A01 Admission message about this encounter. The "full, legal patient name" and birth date is included due to state required reporting of all patients, which is not always the case across all jurisdictions.

Comments

The dates and times in this test case illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test case with EHR technology, only date and time format will be validated within tester submitted test data.

Pre-condition

No Pre-Condition

Post-Condition

No Post-Condition

Test Objectives

This test step examines a Health IT Module's ability to create ADT^A01 Admission message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

- Setting is inpatient.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient Name is Identifiable in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and pseudonymized data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in an identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). SNOMED-CT® codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Admitting Diagnoses
- Multiple Races

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R062	Wheezing	I10C	R062^Wheezing^I10C	Yes
56018004	Wheezing (finding)	SCT	56018004^Wheezing (finding)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R06.2	Wheezing	I10C	R06.2^Wheezing^I10C	Yes
R062	Wheezing	I10	R062^Wheezing^I10	No
R06.2	Wheezing	I10	R06.2^Wheezing^I10	No
785744001	Bronchitis co-occurrent with acute wheeze (disorder)	SCT	785744001^Bronchitis co-occurrent with acute wheeze (disorder)^SCT	Yes
DG1(2)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R509	Fever, unspecified	I10C	R509^Fever, unspecified^I10C	Yes
416113008	Febrile disorder (disorder)	SCT	416113008^Febrile disorder (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R50.9	Fever, unspecified	I10C	R50.9^Fever, unspecified^I10C	Yes
R509	Fever, unspecified	I10	R509^Fever, unspecified^I10	No
R50.9	Fever, unspecified	I10	R50.9^Fever, unspecified^I10	No
DG1(3)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R0681	Apnea, not elsewhere classified	I10C	R0681^Apnea, not elsewhere classified^I10C	Yes
724229002	Apnea of infancy (disorder)	SCT	724229002^Apnea of infancy (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R06.81	Apnea, not elsewhere classified	I10C	R06.81^Apnea, not elsewhere classified^I10C	Yes
R0681	Apnea, not elsewhere classified	I10	R0681^Apnea, not elsewhere classified^I10	No
R06.81	Apnea, not elsewhere classified	I10	R06.81^Apnea, not elsewhere classified^I10	No
1023001	Apnea (finding)	SCT	1023001^Apnea (finding)^SCT	Yes
DG1(4)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R630	Anorexia	I10C	R630^Anorexia^I10C	Yes

700213005	Avoidant restrictive food intake disorder (disorder)	SCT	700213005^ Avoidant restrictive food intake disorder (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R63.0	Anorexia	I10C	R63.0^Anorexia^I10C	Yes
R630	Anorexia	I10	R630^Anorexia^I10	No
R63.0	Anorexia	I10	R63.0^Anorexia^I10	No
192016008	Non-organic infant feeding disturbance (disorder)	SCT	192016008^Non-organic infant feeding disturbance (disorder)^SCT	Yes
DG1(5)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R051	Acute cough	I10C	R051^Acute cough^I10C	Yes
417850002	Respiratory tract congestion and cough (disorder)	SCT	700213005^ Respiratory tract congestion and cough (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R05.1	Anorexia	I10C	R05.1^Acute cough^I10C	Yes
R051	Anorexia	I10	R051^Acute cough^I10	No
R05.1	Anorexia	I10	R05.1^Acute cough^I10	No
49727002	Cough (finding)	SCT	49727002^Cough (finding)^SCT	Yes

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

Codes are provided in the test data for coding the diagnoses and procedures. The Tool is designed to accept those codes without generating an error. If the HIT developer uses a different but equivalent coded value within the same code system than the ones provided/accepted, the Tester may ignore errors generated by the Test Tool related to incorrect code/name of coding system when the coded data used in the message are determined via visual inspection to be valid data.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.